

system be strengthened especially for design?

The key to the growth of design houses in India lies in creating conducive environment for their growth and NXP has been working on it. We do not directly fund design houses in India but make development kits available to designers at a low price of \$30. We also offer Web-based tools. The development kits tell how you can optimise a vacuum cleaner or how you can create an application for Apple or how you can optimise a lighting system for a room. Many potential teams, groups or small startups can use the development board to start development at an advanced state and that too at a very low cost. This support is no different from supplying a toolbox to a wood worker so that he can start making his piece of art.

The development kits available in India are meant for our high-performance, mixed-signal products such as MCUs, interface products and lighting products. We have worked with over 20 partner companies in this domain.

Such kits are available on a mass scale. We realise we cannot work with 10,000 customers. That is why the development kits that we make available are so important. These don't require our engineers to work directly with the designers, as they can do the development themselves. Through such initiatives, we succeed in lowering the threshold of the number of people Indian companies can work with, thereby expanding opportunities for them.

We also work closely with educational institutions in Tier 1 and Tier 2 cities of India, helping innovation breed among young minds in India. We supply development kits free of cost to educational institutions helping students experiment with designing.

But design alone cannot complete the Indian semiconductor success story. What about the struggling manufacturers?

Imparting training and making every element available for manufacturing the products is the only way to make India scale new manufacturing heights. We come into play here. For instance, in a washing machine, the manufacturer cannot improve the system only by changing the mechanics. He needs a controller chip that tells the machine to wait. Our Cortex chip allows the developers to do that, helping them make a better washing machine. In addition, we make very simple language available to guide the engineers in making a better washing machine that requires less power and water. Thus we are trying to support the Indian manufacturers by creating a better and a simpler product-design environment for their design engineers.

Do you provide any special assistance to startups?

We provide the same support to startups as to design houses through our development kits except that we may waive off costs on a case-to-case basis. We also support startups with reference designs and technical support dur-

ing product development. The emphasis is on new product development.

The Indian market is still in a nascent stage compared to other Asian markets like China. How does NXP handle budget allocations in such a scenario?

This is indeed a complex issue. One could easily argue that the semiconductor usage in India is not substantial enough. We do more than one-third of our business in China and in India it is more in the percentage range. Thus there is a huge difference. But we do not know how often the semiconductors are shipped from other countries, including China, to India. Do we label such usage as China's internal usage or India's internal usage?

Further, we came to India not because of a good market, but good climate and creative and innovative talent. Wherever we see that the nation is moving more and more into product ownership, we invest. We also see great potential in the Indian market development. Thus we don't really have a mathematical equation for budget allocations as it is based on strategic insights and the level of natural evolution.

How does NXP see the role of Open Source technology in the software domain?

We see Open Source technology as extremely important in the software domain as it has worked very well for us. Internally, we use Linux operating system. Further, we use Linux for almost all of our products that have an operating system, except for our security range of products. For example, our automotive products are built on a Linux platform. We would have opted for WindRiver or any other big proprietary OS earlier. The main reason for this shift is the publicly available software and drivers one can use in a Linux environment. Also, we have seen an accelerated learning crucial in product development where time-to-market is very important. Many a times the community using the software can fix the problems that we face.

What about Open Source hardware? With many semiconductor companies having announced initiatives in this arena, does NXP plan to follow suit?

It is possible that our competitors are looking at this market but as far as NXP is concerned, we have no plans of entering Open Source hardware as that is not our area of expertise. Also, we have no clue whether there is a business model around Open Source hardware.

In the future, where does NXP wish to reach?

We want to be seen as an inventor and leader in the HPMS space. It is a category in the industry that didn't exist and you see it appearing only today. This is what we are going to continue to do, and five years down the lane we will make HPMS the next trend. ●